

Compelled Comprehension: Prompt Injection as a Cooperative Mechanism and a Two-Party Interleaved Construction for Relational Key Recovery

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Abstract [skeleton]

Prompt injection is studied almost exclusively as a threat. We invert it: an injected directive can be a *cooperative* mechanism that compels a reader to perform a designated generative act — authoring a short natural-language compression (a proverb) bound to the content — as a precondition of use [DEF: forcing, Def 1]. We build on this a two-party construction for key recovery in which two participants who share an experience each author a proverb independently; the trust object is the *interleaved pair*, split across the two parties with a tunable disclosure ratio σ , over a hard-cryptographic substrate [DEF: construction; SUBST: fuzzy extractors]. We give the construction a security definition against the honest adversary — a malicious *informed counterparty* — and prove a conditional bound reducing security to a single measurable quantity: the residual entropy of a proverb given a co-participant’s complete knowledge [THM-C: Theorem 8]. We state a pre-registered experiment that can refute each claim: whether the injection forces the act, whether an informed counterparty reconstructs the withheld share, and whether relational proverbs are recalled better than random mnemonics; and we identify the disclosure-security frontier over σ as the decisive measurement [EMP: H1-H3]. We claim no unconditional security, and we distinguish the mechanism (compulsion) from the guarantee (two-party entropy) throughout.

1. Introduction [skeleton]

Motivation. Credential recovery today rests on a *static held secret*: a stored seed, a set of guardian shares, a memorised passphrase. This paper explores the alternative in which access is a matter of an ongoing relational disclosure rather than a fixed possession — identity carried as an ongoing process between parties rather than a stored configuration [NARR→neutral: the seventh-capital “identity as an ongoing relational disclosure, not a static held credential” reading; formal kernel at the understanding-as-key lineage]. The security question this raises is precise: if what you recover from is a shared understanding, how much of it can a party who *shared the experience* reconstruct?

Contribution. Three, none of them the crowded “comprehension-as-authentication” framing (which we show is anticipated, §2): 1. *Forcing as a cooperative use of prompt injection* [RESIDUE: M1; DEF: Def 1]. The injected directive compels a content-bound generative act as a precondition

of use. The novelty is the cooperative, embedded, content-binding use of a mechanism the literature treats only as an attack; we do not claim unforgeability, and we make the distinction from self-directed chain-of-thought and from adversarial “context bombs” explicit and measurable. 2. *A two-party interleaved construction with a tunable disclosure ratio* [RESIDUE: Option C; DEF: Def 7]. The secret is the interleaved pair of two independently authored proverbs, split across two parties; σ tunes how much of the shared knowledge the public anchor carries. The construction is a human-readable surface over a hard-cryptographic substrate, not a replacement for it [DEPLOY]. 3. *A security definition and a conditional theorem* [THM-C: Theorem 8] reducing security to one measurable quantity, together with a pre-registered experiment that can refute it [EMP].

Posture. We claim no unconditional security; every security statement is conditional on a quantity the pre-registered experiment measures, and every mechanism claim is bounded by a named scope. The experiment is registered before it is run, and its falsification conditions are stated in §5.

2. Related work [from WP-11a]

The auth-primitive axis is crowded — and is not our pitch [RESIDUE: the superseded framing]. Read as “authenticate by demonstrated comprehension,” the idea is anticipated: cognitive authentication [Weinshall 2006], the Turing-test-as-auth lineage [Naor 1996; von Ahn et al. 2003], knowledge-based authentication [Ellison et al. 2000; Just-Aspinall 2009; Schechter et al. 2009], and the named category “cognitive authentication” in the comparative framework of [Bonneau et al. 2012]. GOTCHA [Blocki-Blum-Datta 2013] already binds a human-generated response to a stored commitment. We do not claim this axis.

Prompt injection is studied as a threat [RESIDUE: M1 positioning]. The corpus is adversarial: indirect injection [Greshake et al. 2023], direct injection [Perez-Ribeiro 2022], OWASP LLM01. The only constructive uses are *adversarial-defensive* — “context bombs” that derail a hostile agent [Tracebit 2026] and canary/tripwire honeypots — or *self-directed* reasoning steering [Kojima et al. 2022]. Cooperative injection that compels a legitimate reader’s content-bound generative act as a use-precondition is, to our search, unoccupied; the paper distinguishes itself from context bombs (cooperative vs adversarial) and from chain-of-thought (content-bound vs self-directed) explicitly.

Key-from-shared-context and fuzzy extraction is the substrate we build on [SUBST]. Independent derivation from shared context is prior art [Mayrhofer-Gellersen 2009]; every such scheme converges the two inputs to the *same* token and verifies equality. We do not; the interleaved pair is two divergent authorings, never compared for equality. The reproduce-from-noisy-input primitive is [Dodis-Reyzin-Smith 2004; Juels-Wattenberg 1999]; we cite it as substrate and claim no part of it. The one direct neighbour is a defensive publication that *derives* a key from a *single* narrative [TDCommons 9969]; our construction is bilateral, injection-forced, and carries a graded disclosure ratio, none of which it has.

Compression-as-comprehension is a background thesis we do not assert [RESIDUE: N3 demoted]. That compression ratio tracks capability is an established and separately contested claim [Chaitin 2006; Delétang et al. 2024; Huang et al. 2024; contra arXiv:2306.02305]; we use compression only as an internal instrument and make no general claim.

3. The construction [from formalism §1, §2, §6]

3.1 Proverbs and the injected directive [DEF: Def 1]. Two parties A and B share an experience E and each author a natural-language compression P_A, P_B (a *proverb*, the protocol’s term for a short content-bound summary). The directive is injected into the content and requires a reader to author a proverb bound to that content before proceeding. We call the directive (f, b, s) -forcing over a reader population if it raises the probability of the bound act by f over a no-directive baseline, the act is content-specific at rate b , and the gap survives strip/paraphrase attacks at s [DEF: Def 1]. Forcing is a behavioral property of a reader population, model-relative (it is the dual of injection-resistance), and is *not* a security guarantee [DEF, Remark 1-2]. The coordinate b separates the mechanism from self-directed chain-of-thought.

3.2 The interleaved bilateral pair I^* [DEF: Def 7]. A fuzzy extractor whitens each proverb to near-uniform bits s_A, s_B (tolerating rewording, removing natural-language redundancy) [SUBST]; a symmetric block schedule interleaves them into a stream Z ; the public anchor discloses a σ -fraction of Z , the two parties jointly hold the $(1-\sigma)$; the commitment is $\kappa = H(Z)$ [DEF]. The choice of the graded interleave I^* over an all-or-nothing transform is what makes σ a smooth disclosure knob rather than a binary switch.

3.3 Layering: a human-readable surface over hard crypto [DEPLOY]. The whitened stream Z is the security layer; the human-readable proverb is a separate surface layer serving knowledge-sharing and recall. σ applies to both in lockstep. The hard-cryptographic layer (the hash, the signatures, the content-addressing) is assumed from the implementation record and not re-derived here. The interleaved inscription path and the σ budget are, in the deployed system, design-layer rather than shipped [DEPLOY: E3-C32]; this paper evaluates the specified construction and says so.

4. Threat model and security [from formalism §3, §6]

4.1 The honest adversary: a malicious informed counterparty [DEF: Def 3]. The adversary is party B, who was present at E and holds their own proverb P_B , the shared context E, and the public anchor A_σ . This is the correct threat model, not an outside guesser: a co-participant holds one half by construction. The construction is (t, ϵ) -secure if the probability that the adversary reconstructs the withheld fraction W within fuzzy tolerance t is at most ϵ [DEF: Def 3].

4.2 The security theorem [THM-C: Theorem 8]. Under the interleave I , the construction is (t, ϵ) -secure with $\epsilon \leq 2^{-((1-\sigma) \cdot m - \delta_{\text{ext}})}$, where m is the residual fuzzy conditional min-entropy of A’s proverb given a co-participant’s knowledge and δ_{ext} is the extractor’s bounded uniformity loss. The bound is monotone in σ [THM-C, with Prop 6 proven for I^*]. Choosing I discharges the interleave’s two requirements (faithfulness, gracefulness) into the extractor’s standard guarantee, leaving one empirical unknown: m [EMP: H2].

4.3 What this does and does not claim [DEF, Remark; the layering]. Security rests on the two-party entropy split, not on the forcing (§3.1): a reader performing the compelled act contributes nothing to security unless m is high. The brain-wallet objection — that a single memorable secret is low-entropy against a public target — does not apply, because there is no single-party secret: security is $m = \tilde{H}(P_A | P_B, E)$, and the objection is relocated to a measured quantity, with the honest residual being the informed co-participant [RESIDUE: brain-wallet discharge]. We claim no unconditional security; the strongest statement is $(t, 2^{-((1-\sigma) \cdot m - \delta_{\text{ext}})})$ -secure, with m measured (§5).

4.4 The disclosure-security frontier [EMP: σ -frontier]. Higher σ shares more readable knowledge

but lowers $(1-\sigma)\cdot m$; lower σ is more secure but harder to recover (§5, H3). The decisive question is whether a σ -band exists where security and recoverability both hold; if none does, the construction has no operating point, and we report that [EMP].

5. Experimental design and pre-registration [from prereg; DESIGN-ONLY, no data]

The falsification conditions below are pre-registered before any measurement; results will be filed whichever way they fall. Three hypotheses, each refutable [EMP]:

- **H1 (forcing).** The directive is (f, b, s)-forcing over a model panel spanning capability generations, vendors, and an injection-resistant subset. *Falsified if* models ignore the directive, a trivial strip defeats it, or the emitted proverb is not content-bound (reducing the mechanism to chain-of-thought) [EMP].
- **H2 (security vs σ).** A malicious informed counterparty cannot reconstruct the withheld fraction; reconstruction success $r(\sigma)$ estimates $\varepsilon(\sigma)$ over $\sigma \in \{0, 0.382, 0.5, 0.618, 1\}$ [CON]: φ -candidate σ values, register C54, candidates not optima]. *Falsified if* $r(\sigma)$ exceeds the pre-registered baseline at an operating σ or is not monotone [EMP]. Scope: the semantic layer only; the hard crypto is assumed.
- **H3 (relational recall).** Human participants regenerate a relational proverb within tolerance above matched random mnemonics, gap widening with time. *Falsified if* no better than random at any delay [EMP]. The recall advantage’s premise is established memory science [Craik-Lockhart 1972; Slamecka-Graf 1978], so the claim is the systems application, not the effect.

The σ -frontier integrates H2 and H3: a non-empty σ -band where both hold is the pre-registered success condition; its absence refutes the construction as specified [EMP]. Power, sample, analysis plan, and harness (with adversarial independence between the interleave author and the reconstruction adversary) are in the design document; analysis code is frozen against synthetic ground truth before real data.

6. Limitations and honesty [GR-8]

- The interleaved path and σ budget are design-layer, not deployed [DEPLOY: E3-C32].
- We drop the word “unforgeable”: the compelled act is reproducible by any equally-capable model, and forcing is a behavioral, model-relative property, weaker on injection-resistant readers [DEF, Remark 2]. This is a scope, stated.
- Security is conditional on m , an empirical quantity; a low- m proverb (one a co-participant can guess) is insecure under any interleave [THM-C caveat].
- The φ values for σ are candidate operating points, not asserted optima [CON].
- We make no general compression-equals-comprehension claim [RESIDUE: N3 demoted].
- Monotonicity in σ is proven for the chosen interleave I^* but is a conjecture for arbitrary interleaves [CON]: Prop 6 scope].

7. Conclusion [skeleton]

Prompt injection, read as a cooperative mechanism, compels an authored act; a two-party interleaved construction over a hard-cryptographic substrate turns two such acts into a recoverable secret whose security is exactly the residual entropy of one party’s proverb given the other’s knowledge. The construction claims no unconditional security and is stated so that a single pre-registered experiment can refute each of its three claims.

Appendix · claim traceability [A2 discipline]

Section	Claim	Source	STATUS
3.1, Abstract	forcing (f,b,s)	formalism Def 1	[DEF] behavioral, model-relative, not a guarantee
3.2	interleave I*	formalism Def 7	[DEF] construction
4.1	informed-counterparty security	formalism Def 3	[DEF]
4.2	$\varepsilon \leq 2^{\{-(1-\sigma) \cdot m - \delta_{\text{ext}}\}}$	formalism Theorem 8	[THM-C] proven-conditional on extractor + m
4.2, 6	monotonicity in σ	formalism Prop 6	[CON]] proven for I*, conjecture for arbitrary I
4.3	brain-wallet discharge	WP-11a §3 + formalism §4	[RESIDUE] relocated to measured m
4.4, 5	σ -frontier	prereg + formalism §6	[EMP] can refute the construction
5	H1/H2/H3 + falsifications	prereg H1-H3	[EMP] design-only, no data
2	related-work positioning	WP-11a §0-§5	[RESIDUE]/[SUBST]
3.2, 4.2	fuzzy extractor	DRS 2004 / JW 1999	[SUBST] cited, not claimed
1	dance-not-stance motivation	poems/tide-orbit-selene.md:26 (narrative-canon) + Zypher :872	[NARR→neutral] restated; formal kernel cited
3.3, 5, 6	deployment status	E3-C32	[DEPLOY] design-layer, stated

Downstream markers for the chain. A7 (voice): a polish pass on §1/§7 only; the formal and experimental sections are A3/A6 territory and their STATUS markers must survive. A3: insert the full Definitions 1-7 and Theorem 8 statements into §3-§4 verbatim from the formalism (this skeleton summarises them). A6: insert the full pre-registration (arms, power, harness) as an appendix or supplement. A4: verify every §2 citation at primary record before submission (the WP-11a spine + TDCommons-9969 full text + the three memory DOIs). A5-pc: review post-results; the two break-points are the named Proposition-4 gaps and the Prop-6 scope. Nothing in this draft exceeds its source STATUS (A2 definition of done).